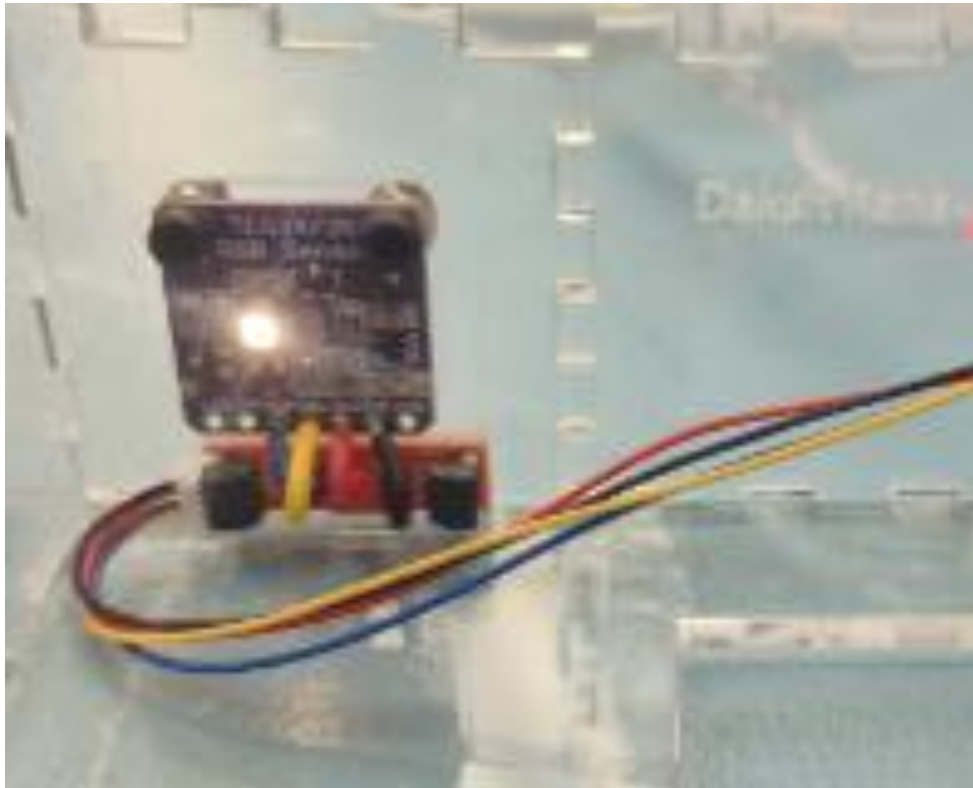




Hardware Production Tech- TCS34725

DAKSH RANA - N01664095

Mounted sensor & i2c Address:-



```
pi@raspberrypi:~ $ sudo i2cdetect -y 1
    0 1 2 3 4 5 6 7 8 9 a b c d e f
00:                .. .. .. .. ..
10: -- .. .. .. .. .. .. .. 1c -- .. ..
20: -- .. .. .. .. .. .. .. 29 .. .. ..
30: -- .. .. .. .. .. .. .. .. .. .. ..
40: -- .. .. .. .. UU .. .. .. .. ..
50: -- .. .. .. .. .. .. .. .. 5c .. .. 5f
60: -- .. .. .. .. .. .. .. 6a .. .. ..
70: -- .. .. .. .. .. .. ..
```

Readings:-

```
Detected Color: GREEN
-----
R: 268, G: 360, B: 231, Clear: 912
Detected Color: GREEN
-----
R: 266, G: 358, B: 230, Clear: 907
Detected Color: GREEN
-----
R: 266, G: 358, B: 230, Clear: 907
Detected Color: GREEN
-----
R: 267, G: 358, B: 230, Clear: 908
Detected Color: GREEN
-----
R: 179, G: 168, B: 108, Clear: 471
Detected Color: RED
-----
R: 1372, G: 824, B: 492, Clear: 2710
Detected Color: RED
-----
R: 1410, G: 584, B: 362, Clear: 2259
Detected Color: RED
-----
R: 1415, G: 631, B: 391, Clear: 2353
```

Sensor Name: TCS34725 RGB Color Sensor
I2C Address: 0x29
Communication: I²C Bus 1
Function: Detects Red, Green, Blue, and Clear light levels to determine object color.

TCS34725 WORKS VIA I²C AT 0X29.

RGB + CLEAR READINGS UPDATE CORRECTLY.

COLOR DETECTION IDENTIFIES RED AND GREEN.

SENSOR IS SECURELY MOUNTED WITH LED ACTIVE.